

Hybrid Machine Learning Approaches for Anti Money Laundering Detection

Code Link: -

<https://gitlab.com/computing.dcu.ie/khushboo.kumari3/hybrid-machine-learning-approaches-for-anti-money-laundering-detection.git>

Contribution & Roles: -

Member	Role	Description of Contribution
Arin Lale	Anomaly Detection, Evaluation & Visualisation	Implemented different techniques to detect anomaly
Gaurang Kumbhar	EDA And Model Evaluation	Performed the EDA on the dataset to get Laundering values and also cleaning the missing values.
Khushboo Kumari	XGBOOST And Random Forest	Implemented all supervised machine learning models like XgBoost and Random forest.
Harshit Mahori	Ensemble Voting, Isolation forest and Performance metrics	Implemented the isolation forest and combined XGboost and Random forest to get the Best accuracy and performance metrics.

Abstract: -

This research proposes a very tough problem of identifying money laundering using highly imbalanced financial data by implementing hybrid machine-learning systems. Since illegal transactions represent less than one percent of the entire activity, the need for identification to both comply with the regulations and efficiently prevent the crimes is necessary. Instead of using one detection paradigm, we perform supervised

learning algorithms, namely XGBoost and Random Forest, along with an unsupervised-based anomaly detection model, which is the Isolation Forest.

Problem And Data: -

Regulatory bodies struggle to identify money laundering among a huge amount of transactions, where fraudulent behaviour constitutes less than 1% of the incidents. To study this phenomenon, we analysed 10000 transactions in various currencies and banks and found only 81 cases (0.81%) of known laundering. The information includes the size of transactions, method of payment, the currency used and the geographical location of the senders and receivers, plus the time.

Methods: -

A combination of preprocessing, intentional design of features, and a hybrid machine-learning pipeline has been our approach to address the overwhelming class imbalance of anti-money-laundering (AML) detection. Categorical variables of type of payment, currency, and banking location will be converted into numbers using LabelEncoder, and continuous variables will be homogenised using StandardScaler to standardise the scales of features. In order to build behavioural patterns, we build both temporal descriptors, such as hour of transaction, day of week, and weekend indicator, and financial ones, including log-transformed amounts of transactions, round amount flags and currency match indicators, all of which indicate some structured or anomalous activity.

Handling Class Imbalance: Since fraudulent laundering instances constitute a minor fraction of transactions, conventional accuracy checks are a misleading move. We only use the SMOTE on the training subset (8,000 samples) to replicate the minority class samples, thus equalising the classes without changing the actual test conditions. An 80/20 split is a stratified sample to ensure that the classes in the training sample are evenly represented.

Model Selection and Training: We evaluate three supervised estimators on the SMOTE balanced data: XGBoost (200 trees, depth 6), Random Forest (200 trees, depth 15) and a soft voting ensemble that combines both models. To do the anomaly detection, we use Isolation Forest (100 estimators), which is trained on valid transactions alone, with isolation-based scoring, which will reveal new suspicious patterns that would otherwise be overlooked by supervised methods.

Evaluation Strategy: We do not rely on accuracy but instead focus on recall and F1-score based on precision-recall curves, ROC-AUC and confusion matrices, which are important measures of minority-class identification. The test set is still biased, and this reflects operational deployments where the cost of not detecting a laundering case is lower than that of false positives.

Results: -

Our hybrid approach combines both supervised and unsupervised education models to deal with the large class imbalance that has defined anti-money laundering (AML) detection problems. The original test set that was inherently imbalanced was used to determine model performance in order to simulate the real-world conditions of deployment, where it is the identification of illicit laundering transactions that is of utmost importance.

Baseline and Model Performance:

A simple classification method was used to determine the baseline. XGBoost and Random Forest, which we supervised, had a recall on the minority-class of around 25 per cent. and an overall recall of around 86 per cent. on the imbalanced test set. This increased accuracy, though, is possibly deceptive with the excessive imbalance in classes because much of it is based on true identification of legitimate transactions. Even though both models effectively modelled the trends in SMOTE-balanced training data, they had a lower ability to maintain sensitivity when faced with naturally small imbalance distributions.

Model Rankings by Performance:

The Ensemble Voting Classifier (Best F1-Score: 0.0299) was the best model, returning an accuracy of 87.0% and a recall of 25 and a precision of 1.59. Its soft-voting algorithm that incorporates XGBoost and random forest improved predictive stability with no deterrent action in minority categories and class. Random Forest (F1: 0.0289) came next with an accuracy of 86.55 per cent with the same recall of 25 per cent, and thus exhibiting a strong tree-based learning on engineered features. The accuracy of XGBoost (F1: 0.0285) in tabular AML data was found to be 86.35 per cent with similar recall, which validates the effectiveness of gradient-boosting methods in tabular data. Isolation Forest (F1: - 0.0173) had the best recall in the minority-class, 62.5 per cent.

Overall Comparison:

It is complemented by the fact that supervised and unsupervised methods are complementary to each other: supervised models are better at detecting patterns of learning with a controlled false alarm rate, whereas Irregular behaviours found by Isolation Forest do not appear in training labels. This two-sidedness justifies our hybrid approach, in which supervised classifiers concentrate on constructed typologies of laundering and anomaly detection identifies new threats, thus building up a powerful, multi-layered AML detection structure.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost	0.8635	0.0151	0.2500	0.0285	0.5165
Random Forest	0.8655	0.0153	0.2500	0.0289	0.5127
Ensemble	0.8700	0.0159	0.2500	0.0299	0.4944
Isolation Forest	0.4325	0.0088	0.6250	0.0173	0.5106

Analysis & Discussion: -

The trade-off critical to the detection of AML is highlighted by our hybrid approach. Supervised models (XGBoost, Random Forest, Ensemble) provide high-quality accuracy (86.35-87 per cent) but detect only a quarter of the cases of laundering- this is a limitation that can be attributed to the fact that SMOTE is not very effective in creating truly representative patterns of minorities in naturally skewed test data. On the contrary, the Isolation Forest recall is 62.5 per cent but generates a high number of false positives, which is inherent to unsupervised anomaly detection and hence necessary to pay a high operational cost unless it is used in conjunction with filtering rules.

Ethical and Legal Considerations: -

AML systems are bound to have compliance with data privacy laws like GDPR and financial compliance laws that regulate the monitoring of transactions. In order to have model transparency, financial institutions should be in a position to explain flagged transactions to the regulators through interpretable models like SHAP or LIME analysis. The privacy risks associated with using synthetic data in research are reduced, but in practice, implementing artificial intelligence would require strong anti-bias and anti-false-positive measures as otherwise, they would entail substantial costs to compliance teams and reduce the detection effectiveness.

Conclusion & Future Work: -

This paper shows that a combination of supervised machine learning and unsupervised anomaly detection provides a more resilient anti-money laundering framework than either of the two methods used alone. XGBoost and Random Forest (our supervised models) could achieve a recall of 25 per cent on data that was critically imbalanced, but Isolation Forest significantly outperformed both of our supervised models because it managed to get 62.5 per cent of recalls on the critically imbalanced data and because it managed to identify interesting patterns that other standard classifiers fail to discover. The Ensemble Voting Classifier also improved the accuracy to 87.0, thus boosting the stability of operation. Future studies ought

to consider models that compute category-specific models, including CatBoost, that potentially better fit categorical attributes that exist in transactional data. Lastly, SHAP-based explainability, which would be implemented, would make it easier to comply with the regulation and provide explanations of risks at the level of a transaction, which is a condition of real-world use.

References: -

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